



CSE422 Lab Project Report

**PCOS Risk Predictor: Identifying At-Risk Individuals
(Group - 20)**

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Introduction:

This project aims to detect the at risk PCOS patients for better healthcare service and load balancing for healthcare professionals. Patients at high risk after doing the tests can be given extra care and can be sent to their desired doctors before the low risk patients for assessment. This will make sure people who need the emergency attention of the doctors get it. PCOS in Bangladesh is still avoided by many women because of a lot of reasons and our social culture is one of them. If this can be implemented just after the small blood test they do. They can be assured by a structured system that they need to visit a specialist as soon as possible.

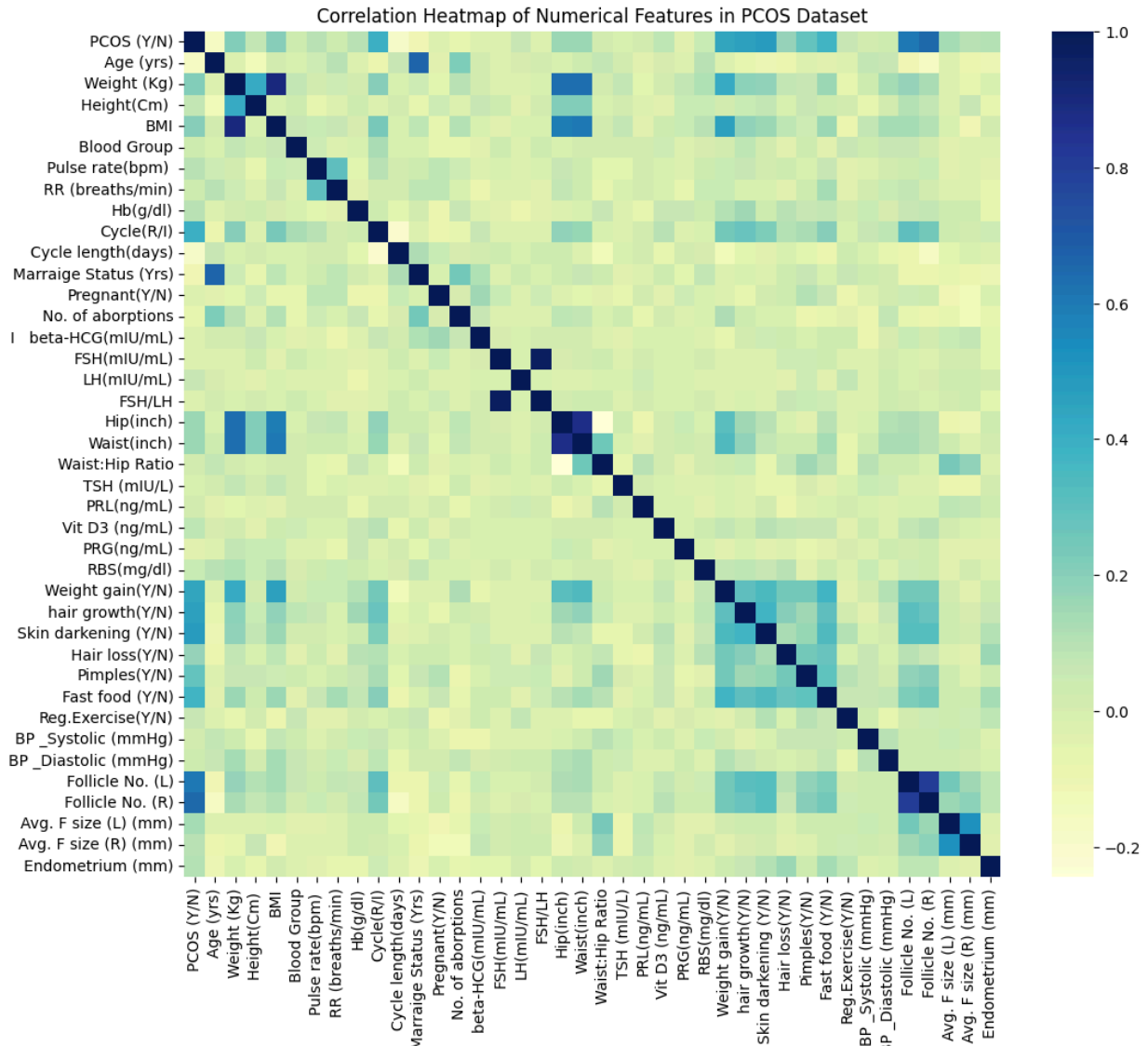
Dataset Description:

Source:

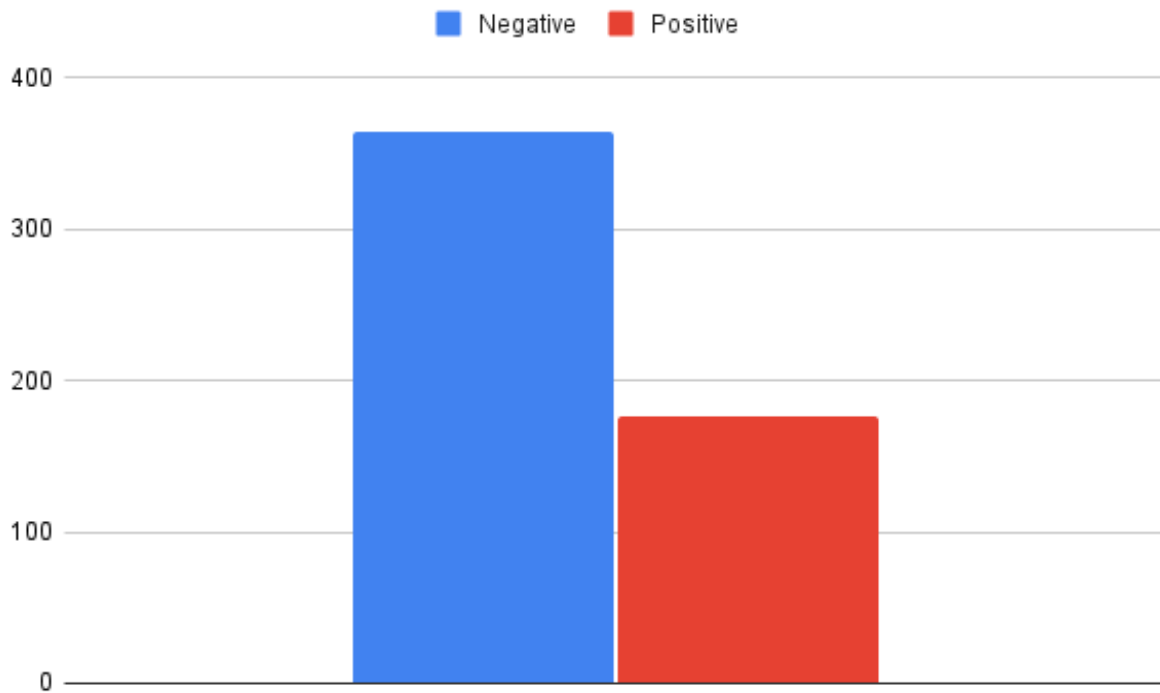
The dataset was collected after going through different mediums and finally found from kaggle (<https://www.kaggle.com/datasets/prasoonkottarathil/polycystic-ovary-syndrome-pcos/data>) suitable for this project. It contains all physical and clinical parameters to determine PCOS and infertility related issues. Data is collected from 10 different hospitals across Kerala, India. Being our neighboring country with similar geographical, environmental and socio-economical similarities, it can be used for our use case too.

The dataset initially contained 44 features with 541 data points. It is a classic classification problem because our target variable is either being PCOS positive or negative. This dataset contained quantitative data with all numerical values.

Correlation of all the features:



Dataset Balance:



There was a slight imbalance between the target variables as negative cases were substantially more than positive cases but it was handled later on by verifying the model not giving false negative results with Confusion Matrix.

Data Pre-Processing:

Faults and Solutions:

There were null or NaN values at 3 places. Firstly, it was how many years a patient was married for. As it was a continuous value, we used mean values of other 540 data points here. Secondly, there was one null value for a YES/NO question for consuming Fast Food. As it was a discrete value, mean was not the solution to use here. So, the most frequent value among all the 540 data points was used here. Thirdly, there was an unnamed column with 539 null values among 541 data points with a mere 2 non null values. As it didn't provide any substantial value for the dataframe, this column was dropped completely.

Feature Scaling:

We did not perform any feature scaling as it was not needed in our case and for this dataset.

Dataset Splitting:

Our dataset was randomly splitted where 378 data points (70%) were used for the Training set and 163 data points (30%) were used for the testing set. It was randomized by random state seed 25.

Model training & testing:

For our classification problem, we decided to use both linear and non linear models and come up with the best performing model.

Linear models:

Logistic Regression:

The accuracy for logistic regression came up to be 89.57% closing to a very acceptable range of results. The other statistics were as follows:

	precision	recall	f1-score	support
0	0.90	0.94	0.92	107
1	0.88	0.80	0.84	56
accuracy			0.90	163
macro avg	0.89	0.87	0.88	163
weighted avg	0.90	0.90	0.89	163
Confusion Matrix:				
[[101 6]				
[11 45]]				

Support Vector Classifier (SVC):

The accuracy for Support Vector Classifier (SVC) came up to be 88.96% closing to a very acceptable range of results. The other statistics were as follows:

	precision	recall	f1-score	support
0	0.89	0.94	0.92	107
1	0.88	0.79	0.83	56
accuracy			0.89	163
macro avg	0.89	0.86	0.87	163
weighted avg	0.89	0.89	0.89	163

SVC Confusion Matrix:

```
[[101  6]
 [ 12 44]]
```

Non Linear Models:

Decision Tree:

The accuracy for Decision Tree came up to be 85.27% closing to an acceptable range of results. The other statistics were as follows:

	precision	recall	f1-score	support
0	0.88	0.90	0.89	107
1	0.80	0.77	0.78	56
accuracy			0.85	163
macro avg	0.84	0.83	0.84	163
weighted avg	0.85	0.85	0.85	163

Decision Tree Confusion Matrix:

```
[[96 11]
 [13 43]]
```

Gradient Boosting:

The accuracy for Gradient Boosting came up to be 90.18% closing to a very acceptable range of results. The other statistics were as follows:

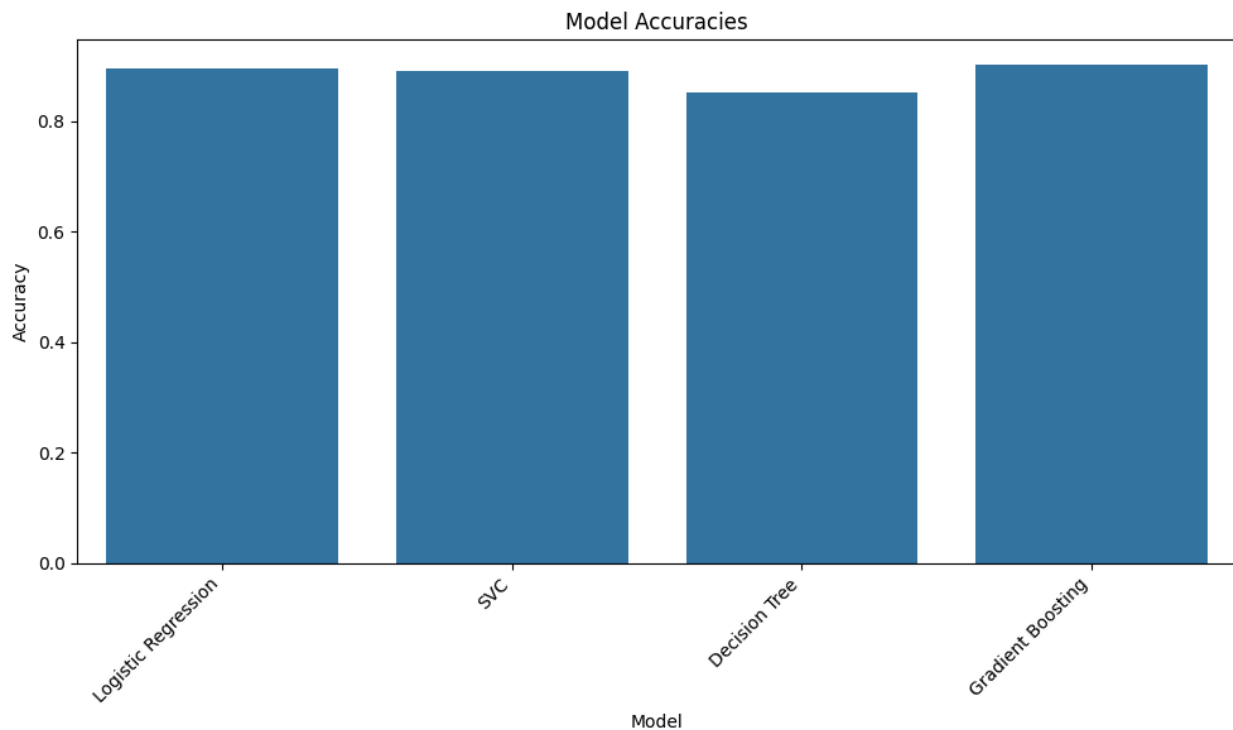
	precision	recall	f1-score	support
0	0.93	0.92	0.92	107
1	0.84	0.88	0.86	56
accuracy			0.90	163
macro avg	0.89	0.90	0.89	163
weighted avg	0.90	0.90	0.90	163

Gradient Boosting Confusion Matrix:

```
[[98  9]
 [ 7 49]]
```

Model Selection/Comparison Analysis:

Among the four trained and tested models, all of them showed very promising results. The results are as follows:



Here, Gradient Boosting is showing the best result with 90.18% accuracy and giving only 9 False negatives. It is also showing better precision and recall results. This can be the best model for selecting for this dataset and this problem.

Conclusion:

This project explains how simple training a model can be for detecting PCOS at an early stage, utilizing a light weight model such as Gradient Boosting which can predict if a patient is at risk of developing PCOS. If implemented correctly it can help a lot of patients to develop future complications such as cervical cancer.